

The Geometry of Nut

The Book of Nut Re-Read Through
the Egyptian Embedding Space

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March 2026

Abstract

The Book of Nut, painted on the ceilings of New Kingdom royal tombs, describes the sun’s journey across the body of the sky goddess—swallowed at dusk, traveling through her body at night, and reborn at dawn. Standard Egyptology reads this as mythological astronomy. This document re-reads the Book of Nut line by line through the Egyptian embedding space—10,833 ancient Egyptian words projected into 300-dimensional GloVe English space via Ridge regression. For each key passage, we compare the standard translation against what the geometry of the language itself reveals. The line-by-line analysis shows that Nut is not sky but time; that the Duat (underworld) is a solar concept, not a chthonic one; that night is a temporal position rather than a visual condition; and that the nocturnal journey of Ra is, in the language’s own geometry, a process of transformation within time, not a transit through darkness.

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1 Introduction

The Book of Nut (*Nut Fundamentaltext*) survives in several New Kingdom royal tombs, most completely in the cenotaph of Seti I at Abydos (c. 1280 BCE) and the tomb of Ramesses IV (KV 2). It describes the mechanics of the cosmos: how the sun enters Nut’s mouth at the western horizon, travels through her body during the twelve hours of night, and is born from her at dawn in the east. The stars follow a similar cycle. The text is accompanied by a monumental image of Nut arched over the earth, her body studded with stars, Shu supporting her from below.

Every translation treats the Book of Nut as a work of mythological astronomy—a pre-scientific account of celestial mechanics dressed in divine narrative. The question this document asks is whether the Egyptian language itself supports this reading, or whether the astronomical framework is a modern projection.

1.1 Method

Each Egyptian word in the key passages was looked up in the V15 aligned embedding space (10,833 words projected into GloVe 300d). For each word, we record the nearest English neighbors and the nearest Egyptian neighbors, identifying where these mappings diverge from the standard translation’s implied framework. The passages below follow the text as reconstructed by [von Lieven \[2007\]](#) and [Neugebauer and Parker \[1960\]](#).

1.2 How to Read the Line-by-Line Analysis

Each passage is presented in three layers:

1. **The Egyptian passage** in transliteration, with the standard English translation.
2. **Word-level geometry**: Key words are looked up in the embedding space. English neighbors (what the word maps to in GloVe space) and Egyptian neighbors (what the word clusters with in the native vocabulary) are reported.
3. **Geometric reading**: What the passage says when the embedding-space meanings replace the conventional translations.

2 Line-by-Line Analysis

2.1 Passage 1: The Figure of Nut

“Nw,t tn — p,t pw ,t”

Standard translation: “This Nut — she is the great sky.”

Nw,t (the goddess Nut)

English neighbors: “time” (0.59), “when” (0.57), “come” (0.55), “before” (0.55)

Egyptian neighbors: *s-Nw,t* (son of Nut, 0.81), *Nw,w* (0.76), *z-Nw,t* (0.76)

p,t (sky)

English neighbors: “sky” (0.86), “skies” (0.51), “blue” (0.45), “night” (0.43), “bright” (0.43)

Egyptian neighbors: *šp,t* (0.69), *ḥp,t* (0.67), *p,t* (0.66)

Geometric reading: The opening line identifies Nut with the sky, but the two words occupy entirely different semantic regions. *P,t* maps cleanly to “sky” (0.86)—spatial, visual, blue, bright. *Nw,t* maps to “time” (0.59)—temporal, sequential, processual. The text says “this Nut is the great sky” but the language says *Nw,t* is not *p,t*. They are geometrically unrelated. Either the text is making an equation the language doesn’t support, or the passage means something closer to: “This Time—she is the great expanse.”

2.2 Passage 2: The Sun Enters Nut’s Mouth

“ m R m r s m jmn,t ”

Standard translation: “Ra enters her mouth in the west.”

m (swallow, enter)

English neighbors: “not” (0.68), “know” (0.67), “if” (0.67)

Egyptian neighbors: *m m* (devour, 0.80), *m* (0.79), *m m* (0.78)

jmn,t (west)

English neighbors: “west” (0.91), “east” (0.78), “western” (0.77), “eastern” (0.67), “north” (0.63)

Egyptian neighbors: *jmn,tt* (0.95), *jmn,tj* (western, 0.93), *jmn,w* (0.92)

Geometric reading: The west (*jmn,t*) maps cleanly—directional, spatial, confirmed at 0.91. But the verb *m* is revealing. Its Egyptian neighbors are all intensified forms of consumption: *m m* (to devour repeatedly), *m m* (to gulp). This is not gentle ingestion. The geometry of the verb field describes violent, total consumption. Ra does not “enter” Nut’s mouth; Ra is *devoured* by time. The western horizon is where time consumes the sun utterly.

2.3 Passage 3: The Hours of Night

“wnw,t 12 n,t grḥ — nm,t f m-ḥnw s”

Standard translation: “The 12 hours of night — he travels inside her.”

wnw,t (hour)

English neighbors: “hour” (0.76), “hours” (0.66), “day” (0.61), “night” (0.60)

Egyptian neighbors: *wnw,t.pl* (hours, 0.86), *nw,t* (0.85), *tnw,t* (0.83)

grḥ (night)

English neighbors: “came” (0.61), “night” (0.59), “last” (0.57), “first” (0.56)

Egyptian neighbors: *grḥ,w* (nights, 0.88), *nw,t.pl* (0.88), *sbḥ,t.pl* (0.87)

nm,t (travel, stride)

English neighbors: “which” (0.73), “where” (0.70), “only” (0.70)

Egyptian neighbors: *knm,t* (0.96), *Ḥnm,t.pl* (0.95), *ssm,t.pl* (0.94)

Geometric reading: Night (*grḥ*) maps to temporal sequence—“came,” “last,” “first”—not to darkness or absence of light. It is a position in a cycle, not a visual condition. The hour (*wnw,t*) maps cleanly to its English equivalent. But note that *wnw,t*’s Egyptian neighbor *nw,t* (0.85) is essentially Nut herself—the hours cluster with the goddess. If Nut is time, then the

hours are not divisions *of* her; they *are* her, particularized. “The 12 hours of night” becomes “the 12 modes of time in the sequential position.” Ra does not travel through a dark body; Ra passes through twelve distinct temporal intensities.

2.4 Passage 4: The Duat

“*wnn f m dw ,t*”

Standard translation: “He is in the Duat (underworld).”

dw ,t (Duat, underworld)

English neighbors: “come” (0.70), “this” (0.65), “next” (0.64)

Egyptian neighbors: *Šd,w-m-dw ,t* (He-who-is-in-the-Duat, 0.92), *dw ,yt* (variant, 0.91)

dw ,w (morning, dawn, tomorrow—etymological relative)

English neighbors: “come” (0.66), “tomorrow” (0.65), “here” (0.63), “go” (0.62), “next” (0.62)

Egyptian neighbors: *dw ,yt* (0.89), *dw (,w)* (0.89), *dw .ww* (0.86)

Geometric reading: The Duat (*dw ,t*) and the morning (*dw ,w*) share their root and their geometry. Both map to English “come” and “next.” The Duat is not an underworld of darkness—it is the *place of the coming thing*, the space of “tomorrow” and “next.” Its Egyptian neighbors confirm: the Duat clusters with its own denizens and variants, all sharing the same dawn-oriented semantic field. The standard translation puts Ra “in the underworld”; the geometry puts Ra *in the place where morning is produced*. The Duat is the workshop of dawn.

2.5 Passage 5: The Horizon

“*prrr f m ḥ,t j b,t,t*”

Standard translation: “He comes forth from the eastern horizon.”

prrr (come forth, emerge)

English neighbors: “which” (0.65), “is” (0.63), “part” (0.59)

Egyptian neighbors: *prrr!* (emphatic, 0.90), *prḥ* (0.89)

ḥ,t (horizon)

English neighbors: “which” (0.63), “part” (0.60), “this” (0.59)

Egyptian neighbors: *jm ḥ,t* (0.95), *nb-ḥpš* (lord of strength, 0.95), *ḥw,t.pl* (temples/estates, 0.95), *nb-tm* (lord of totality, 0.94)

j b,t (east)

English neighbors: “osten” (east [German], 0.67)

Egyptian neighbors: *j b,tj* (eastern, 0.92), *j b,tt* (0.90)

Geometric reading: The horizon (*ḥ,t*) does not map to a geographical feature. Its Egyptian neighbors are *nb-ḥpš* (lord of strength), *nb-tm* (lord of totality), and *ḥw,t.pl* (temples). The horizon is geometrically a zone of *lordship, totality, and sacred architecture*—not a line where earth meets sky, but a threshold of maximum cosmic intensity. Ra “comes forth from the eastern horizon” becomes Ra *emerges from the eastern threshold of total power*. This explains why the horizon-glyph appears in the names of temples and royal monuments: they participate in the horizon’s quality as a zone of absolute force, not its quality as a geographical location.

2.6 Passage 6: Birth at Dawn

“*msj sw Nw,t m dw ,w*”

Standard translation: “Nut gives birth to him in the morning.”

msj (give birth, bear)

English neighbors: “when” (0.69), “he” (0.71), “so” (0.67)

Egyptian neighbors: *msy* (variant, 0.94), *ms.n* (has given birth, 0.93), *msdd* (0.93)

msw,t (birth—nominal form)

English neighbors: “first” (0.68), “when” (0.68), “same” (0.68)

Egyptian neighbors: *hsw,t* (praise/favor, 0.94), *msḥn* (birth-place, 0.91)

dw,w (morning, dawn)

English neighbors: “come” (0.66), “tomorrow” (0.65), “next” (0.62)

Geometric reading: The birth (*msw,t*) neighbors “first” and *hsw,t* (praise/favor). Dawn (*dw,w*) is “come” and “tomorrow.” Together: Nut does not give birth to the sun as a biological act—time *produces* the sun as the “first” and “coming” event. The birth is a temporal emergence. The sun exits the medium of time as a renewed entity. Each morning is not a return but a *first* occurrence—the geometry of *msw,t* neighbors “first,” suggesting that each sunrise is experienced as original, not repeated. Time gives birth, and each birth is new.

2.7 Passage 7: Shu Supports Nut

“*Šw ḥr stz Nw,t*”

Standard translation: “Shu supports/lifts Nut.”

Šw (Shu)

English neighbors: “thing” (0.44), “want” (0.44), “really” (0.42)

Egyptian neighbors: *tm* (0.60), *n,t-* (0.60), *nš* (0.60)

stz (lift, raise, support)

English neighbors: “well” (0.63), “which” (0.61), “this” (0.60)

Egyptian neighbors: *stz,w* (the lifter, 0.97)

stz,w-Šw (the lifting of Shu—compound)

Egyptian neighbors: *stz,w* (0.97)

Also appears as neighbor of *wbn* (sunrise, 0.77)

Geometric reading: Shu (*Šw*) does not map to air or atmosphere in the embedding space—the standard identification of Shu with air has no geometric support. The compound *stz,w-Šw* (the lifting of Shu) appears in the neighborhood of *wbn* (sunrise). Every sunrise recapitulates Shu’s original act. Shu does not hold up the sky; Shu *separates temporal experience from material existence*. If Nut is time and Geb is earth, Shu’s role is to maintain the structural distinction between time and matter—a distinction that collapses each night when Ra is devoured, and that must be re-established each dawn.

2.8 Passage 8: The Stars

“*sb .pl ḥr nm,t ḥr ḥr,t s*”

Standard translation: “The stars travel upon her upper surface.”

sb (star)

Egyptian neighbors: a large cluster of high-similarity terms (many at 1.00), indicating the word sits in a dense, highly-connected region of Egyptian semantic space.

ḥr,t (upper surface, upper region)

English neighbors: “sky” (0.64), “come” (0.63)

Egyptian neighbors: *ḥr,y* (upper, 0.82), *ḥrj* (0.81)

Geometric reading: The word *hr,t* (upper surface) maps to both “sky” (0.64) and “come” (0.63)—spatial and temporal equally. The stars (*sb.pl*) sit in an extremely dense semantic cluster, suggesting they function as fundamental structural units in the Egyptian conceptual vocabulary. If Nut is time, stars traveling on her surface are not astronomical objects moving through the sky but *temporal events* distributed across the medium of time. The “upper surface” of Nut is the *accessible face of temporal experience*—the part of time that can be observed and measured.

2.9 Passage 9: The Stars Enter and Exit

“*m sn m r s m jmn,t — msj s sn m j b,t*”

Standard translation: “They enter her mouth in the west — she gives birth to them in the east.”

This passage applies the same verbs used for Ra’s journey to the stars: they are *devoured* (*m*) in the west and *born* (*msj*) in the east, following the same cycle.

Geometric reading: The stars undergo the same temporal process as the sun. They are consumed by time in the west and produced by time in the east. This is not a description of stellar risings and settings—it is a description of how temporal experience consumes and regenerates all phenomena. Every event is devoured by time and reborn from time. The stars are not special because they are astronomical; they are special because they visibly demonstrate the universal temporal cycle. They are time’s own evidence for its nature.

2.10 Passage 10: The Two Eternities

“*nḥḥ d,t*”

Standard translation: “Cyclical eternity and linear eternity” (often rendered “forever and ever”). Appears throughout the Book of Nut as the temporal frame of Ra’s journey.

nḥḥ (cyclical eternity)

English neighbors: “and” (0.96), “both” (0.72), “well” (0.68), “including” (0.63)

Egyptian neighbors: *hw* (0.96), *nf* (0.96), *hn* (together with, 0.89)

d,t (linear eternity)

English neighbors: “and” (0.86), “both” (0.73), “well” (0.66), “other” (0.65), “also” (0.63)

Egyptian neighbors: *hw* (0.86), *nf* (0.86), *r-d,t* (unto eternity, 0.86)

Geometric reading: Both eternities map to connective, inclusive English—“and,” “both,” “including.” They are not durations but *conjunctions*—ways of holding things together. *Nḥḥ* (cyclical) and *d,t* (linear) share the Egyptian neighbor *hw* (0.96 and 0.86 respectively), suggesting they are two modes of a single underlying connective force. The Book of Nut operates in *nḥḥ*—cyclical recurrence—while the beings within it (the dead, Osiris, the stars that have “died”) persist in *d,t*—unbroken duration. The text describes a cycle that contains permanences: things that endure forever ride within a process that repeats forever. Two different kinds of forever, both required.

2.11 Passage 11: Life and the Living

“*nḥ R — wḥm msw,t f*”

Standard translation: “Ra lives — he repeats his birth.”

nḥ (live, life)

English neighbors: “live” (0.88), “living” (0.60), “where” (0.63), “well” (0.61)

Egyptian neighbors: *d* (duration/eternity, 0.88), *nḥw* (the living, 0.83)

wḥm (repeat, renew)

English neighbors: “first” (0.70), “he” (0.71)

Egyptian neighbors: *wḥm.w* (repeater, 0.95), *P-wḥm* (0.94), *m-wḥm* (in repetition, 0.92)

Geometric reading: Life (*nḥ*) neighbors eternity (*d*, 0.88)—to live is to endure. Repetition (*wḥm*) clusters with its own intensified forms, describing a robust concept of cyclical renewal. “Ra lives—he repeats his birth” becomes: Ra *endures*—he *renews* his temporal emergence. The statement is not biological (Ra is alive) but ontological (Ra persists through the mechanism of cyclical renewal). Life and repetition are the same process viewed from different angles.

3 General Analysis

The line-by-line analysis reveals a systematic pattern: wherever the standard translation assumes spatial-astronomical reference, the embedding space yields temporal-processual meaning. This section synthesizes the findings.

3.1 Nut Is Time, Not Sky

The most consequential finding is the geometric separation between *Nw,t* (time: 0.59) and *p,t* (sky: 0.86). These two words occupy entirely different regions of semantic space. *P,t* maps to spatial-visual concepts—blue, bright, above. *Nw,t* maps to temporal concepts—when, before, come.

This means the Book of Nut is not a text about the sky. It is a text about time. The monumental image of Nut arched over the earth is not a diagram of the physical cosmos; it is a depiction of temporal experience—the surface across which all events unfold. The stars on her body are not objects in a sky; they are events in time.

3.2 The Duat as Dawn-Workshop

The Duat (*dw,t*) shares its root and its geometry with morning (*dw,w*). Both map to “come” and “next.” The Western concept of an underworld as a dark, subterranean realm has no geometric support in Egyptian space. The Duat is a solar concept—the place where dawn is generated. The Book of Nut’s description of Ra traveling through Nut at night is not a journey through darkness but a journey through the *place where morning is produced*.

The Amduat’s description of the nocturnal hours as brightly lit, populated by fire-breathing serpents and blazing lakes, is consistent with this reading. The Duat is not dark because it is not an underworld—it is a solar-temporal workshop.

3.3 Night as Temporal Position

Night (*grḥ*) maps to sequential terms—“came,” “last,” “first”—not to darkness or the absence of light. The Egyptian night is not a visual condition but a position in a cycle. This reframes the entire nocturnal journey: Ra does not pass through darkness but through a sequence of temporal positions, each with its own character and intensity.

3.4 Sunrise as Re-Creation

The compound *stz,w-šw* (the lifting of Shu) appears in the neighborhood of *wbn* (sunrise, 0.77). Every sunrise recapitulates the original creation—the separation of time from earth. The Book of Nut does not describe a daily routine; it describes a daily re-creation. Shu is not atmosphere holding up a sky-vault; Shu is the structural force that maintains the distinction between temporal and material existence. Each dawn, that distinction must be re-established.

3.5 The Horizon as Threshold of Power

The horizon (h, t) neighbors *nb-hpš* (lord of strength, 0.95), *nb-tm* (lord of totality, 0.94), and *hw, t.pl* (temples, 0.95). It is not a geographical feature but a zone of maximum cosmic intensity. This explains why the horizon-glyph is used in the names of temples and royal monuments: they participate in the horizon’s quality as a threshold of total power, not a line on a landscape.

4 Discussion

4.1 The Book of Nut as Temporal Treatise

If the readings above are correct, the Book of Nut is not a work of mythological astronomy. It is a treatise on the nature of time—how time functions, what happens within time, how entities persist and transform within temporal experience, and how the original creation (the separation of time from matter) is renewed with each sunrise.

This reading resolves several long-standing puzzles:

1. **Why is Nut depicted swallowing and birthing the sun?** Because time consumes and regenerates events. The sun does not move through space; it is consumed by time and re-emerged by time.
2. **Why are the decans on Nut’s body?** Because they are temporal events, not astronomical objects. They mark the structure of time, not the positions of stars.
3. **Why is the Duat not dark?** Because it is the place of the dawn—the temporal workshop where morning is generated. The Amduat’s description of the nocturnal hours as brightly lit makes sense if the underworld is a solar-temporal space.
4. **Why does the Book of Nut appear on tomb ceilings?** The dead enter Nut—they enter time. The tomb ceiling is not a sky map but a temporal map, showing the deceased the structure of the eternity they are about to inhabit.

4.2 Implications

The Nut-as-time hypothesis is testable. If correct, passages in the Book of Nut that have resisted coherent spatial-astronomical translation should yield more coherent temporal translations. Passages that describe the “distance” between stars should read better as the “duration” between events. Passages that describe Nut’s body as a “vault” should read better as a “span” (of time). This is a target for future geometric analysis of the full Book of Nut text.

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